

# Winning Probabilities as Credit: Fast, Redundancy-Aware Attribution

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## Abstract

The probability that a competitor wins a noisy race—the Thurstonian choice probability, equivalently the gradient of an expected-maximum potential—is a rule for attributing credit among correlated contributors. It is a probability share ( $\sum_i w_i = 1$ ), symmetric, and *redundancy-aware* (contributors that move together share a single contributor’s credit rather than double-counting); it is differentiable in the contributors’ abilities, it updates online, and it costs  $O(Mn)$  in a single Monte-Carlo pass with no coalition enumeration. It is *not* a Shapley value and does not try to be: Shapley measures coalitional cooperation—a contributor’s average marginal value as coalitions form—whereas the winning probability measures *selection relevance*, how often a contributor is the single best in the field at hand. The two answer different questions, and a contributor that is never individually best yet improves every blend it joins earns Shapley credit but little race credit; the race does not capture such coalitional value. (There is a clean mathematical link—the winning-probability field integrates along the diagonal to an Aumann–Shapley value of an ability-scaling game—which we record but do not lean on.) We show that the headline redundancy property—an even split of credit among near-duplicates—is a property of the *calibrated* equal-ability race rather than of raw scores, and we demonstrate the rule on forecast combination and feature attribution. This develops the credit reading of the construction in the companion paper on Thurstone portfolios.

## 1 Introduction

Attributing credit for a collective outcome among contributors that overlap is an old problem with a canonical answer, the Shapley value (Shapley 1953), and a canonical difficulty, its cost: the exact value sums marginal

contributions over all  $2^n$  coalitions, and the practical estimators sample permutations (Štrumbelj and Kononenko 2014) or, in machine learning, fit a local surrogate (SHAP, Lundberg and Lee (2017)). In finance the same gradient logic appears as Euler risk-capital allocation, where a unit’s charge is its marginal contribution to a coherent risk measure (Denault 2001; Kalkbrenner 2005; Tasche 2008). Correlated contributors compound the difficulty: a contributor and a near-duplicate should *share* credit, not each receive it in full, and many cheap importance measures (marginal screening, regression coefficients, permutation importance) either double-count the pair or split it arbitrarily.

We study a different credit rule with a single, cheap evaluation. Give each contributor  $i$  a latent ability  $\theta_i$  and a noisy performance  $X_i = \theta_i + \eta_i$  under a joint law  $S$ ; the credit of  $i$  is the probability that it wins,

$$w_i = \mathbb{P}(X_i = \max_j X_j). \quad (1)$$

We follow the lowest-*score*-wins convention of the **thurstone** package and Cotton (2021):  $\theta_i$  is a *strength*, the negated score, so the lowest-score contributor is the  $\arg \max$  winner in (1); in the loss-based applications of Section 6 the score is a loss and the lowest-loss contributor wins. This is the Thurstonian choice probability, and  $w = W_S(\theta)$  is the gradient of an expected-maximum potential (Section 2). Read as credit it has three things going for it at once. It is redundancy-aware by construction: two contributors that move together compete with *each other*, so a co-moving pair collapses toward a single winner and shares one contributor’s credit. It is differentiable in  $\theta$ , so it composes with gradient-based learning. And it costs one Monte-Carlo pass,  $O(Mn)$ , returning all  $n$  shares at once with no coalition enumeration.

Our contributions are (i) the precise relationship to the Shapley value: the winning probability is the gradient of the same expected-maximum potential whose *diagonal integral* is an Aumann–Shapley value (Owen 1972; Aumann and Shapley 1974), but it is not itself the Shapley value of the membership game (Section 3); (ii) the computational separation,  $O(Mn)$  versus  $O(2^n)$ , quantified (Section 4); and (iii) the clarification that the headline redundancy property—an *even* split among near-duplicates—belongs to the *calibrated* equal-ability race, not to raw contribution scores (Section 5). We close with two applications, forecast combination and feature attribution (Section 6).

## 2 The credit rule

Let  $S$  be a centered joint law on  $\mathbb{R}^n$  (the *simulation*) and define the expected-maximum potential and its gradient,

$$G_S(\theta) = \mathbb{E}_{\eta \sim S} \left[ \max_i (\theta_i + \eta_i) \right], \quad W_S(\theta) = \nabla G_S(\theta) \in \Delta, \quad (2)$$

where  $\Delta$  is the simplex.  $G_S$  is convex for any  $S$ ; when  $S$  is continuous and the usual dominated-differentiation conditions hold, the maximizer is a.s. unique,  $G_S$  is differentiable, and  $\nabla_i G_S(\theta) = \mathbb{P}(i = \arg \max_j X_j) = w_i$ , recovering (1) (the Williams–Daly–Zachary / perturbed-maximum identity (Williams 1977; McFadden 1981; Berthet et al. 2020)); for Gaussian  $S$ ,  $G_S$  is  $C^\infty$ . Because  $G_S(\theta + c\mathbf{1}) = G_S(\theta) + c$ , gradients and Jacobians live on the tangent space  $T\Delta = \{v : \mathbf{1}^\top v = 0\}$ .

To use (1) as an attribution we must fix the abilities. Given a *benchmark*  $w^0 \in \text{int } \Delta$  of marginal credits and a reference law  $S_0$ , calibration inverts the gradient map,

$$\theta^0 = W_{S_0}^{-1}(w^0) \pmod{\mathbf{1}}, \quad (3)$$

which is well posed: for Gaussian  $S_0 = \mathcal{N}(0, C_0)$  with  $C_0 \succ 0$ ,  $G_{C_0}$  is a Legendre-type convex function on the quotient  $\mathbb{R}^n / \text{span}\{\mathbf{1}\}$ —smooth, strictly convex, and steep toward the boundary of its image—so  $W_{C_0} = \nabla G_{C_0}$  is a smooth bijection onto  $\text{int } \Delta$ ; the diagonal case admits a linear-time, Monte-Carlo-free inverse (the Fast Ability Transform, Cotton (2021)). A zero target weight sends its ability to  $-\infty$ , so such contributors are dropped or floored. The credit rule is then

$$T_{S_0 \rightarrow S_1}(w^0) = W_{S_1}(W_{S_0}^{-1}(w^0)), \quad (4)$$

the winning probabilities of the same abilities re-evaluated under a dependence-rich law  $S_1$ . With  $S_1 = S_0$  the benchmark is reproduced; with  $S_1$  carrying the contributors’ empirical dependence, (4) re-weights the benchmark for redundancy.

## 3 Relationship to the Shapley value: a contrast, not a substitute

The winning probability is naturally compared to the Shapley value, and the relationship is real but easy to overstate, because the two are credit in

different *units*. The winning probabilities are *probability shares*,  $\sum_i w_i = 1$ . The Shapley value allocates a cooperative game's grand value: for

$$v(\mathcal{S}) = \mathbb{E}\left[\max_{i \in \mathcal{S}} X_i\right], \quad v(\emptyset) = 0, \quad (5)$$

with grand value  $v(N) = \mathbb{E}[\max_i X_i] = G_S(\theta)$ , the Shapley shares sum to  $v(N)$ —a payoff in the units of  $X$ , not to one. So  $w$  is *not* a Shapley value of  $v$ , and the two obey different efficiency conditions,  $\sum_i w_i = 1$  versus  $\sum_i \phi_i = v(N)$ . The honest connection is a gradient/integral one, through the ability-scaling game.

**Proposition 1** (Gradient, integral, and the membership value). *Let  $S$  be continuous with the usual dominated-differentiation conditions, and define the ability-scaling game  $f(x) = G_S(x \odot \theta) - G_S(0)$  for  $x \in [0, 1]^n$ , where  $x \odot \theta$  scales ability  $i$  by  $x_i$ .*

- (i) Gradient. *The winning-probability vector is the endpoint gradient,  $w = \nabla G_S(\theta)$ , with  $\sum_i w_i = 1$ .*
- (ii) Aumann–Shapley. *The diagonal Aumann–Shapley value of  $f$  (Aumann and Shapley 1974) is  $AS_i = \int_0^1 \partial_{x_i} f(t\mathbf{1}) dt = \theta_i \int_0^1 W_S(t\theta)_i dt$ , in payoff units; it is efficient,  $\sum_i AS_i = G_S(\theta) - G_S(0)$ , and symmetric. The winning-probability field  $\theta \mapsto W_S(\theta)$  is the gradient whose diagonal line integral (weighted by  $\theta$ ) is  $AS$ ;  $w$  is the endpoint gradient, not the integral.*
- (iii) Membership Shapley. *The Shapley value of (5) is, by Owen's multilinear extension (Owen 1972), the diagonal integral over a random coalition  $T$  of the marginal  $v(T \cup \{i\}) - v(T)$ , which equals  $\mathbb{E}[(X_i - \max_{j \in T} X_j)^+]$  for  $T \neq \emptyset$  and  $\mathbb{E}[X_i]$  for  $T = \emptyset$ . This payoff-valued, coalition-averaged quantity is a different object from the probability  $w_i = \mathbb{P}(X_i = \max_N)$ , and the two generally differ (they agree in the fully symmetric case).*

*Proof.* (i) is the Williams–Daly–Zachary / perturbed-maximum identity (Williams 1977; Berthet et al. 2020), valid under the stated regularity. For (ii), the chain rule and (i) give  $\partial_{x_i} f(x) = \theta_i W_S(x \odot \theta)_i$ ; on the diagonal this is the stated integrand, and efficiency is the fundamental theorem of calculus,  $\sum_i AS_i = \int_0^1 \frac{d}{dt} G_S(t\theta) dt = G_S(\theta) - G_S(0)$ ; symmetry follows because exchanging equal-ability contributors leaves  $G_S$  invariant. For (iii), when  $T \neq \emptyset$ ,  $v(T \cup \{i\}) - v(T) = \mathbb{E}[\max(\max_{j \in T} X_j, X_i)] - \mathbb{E}[\max_{j \in T} X_j] =$

$\mathbb{E}[(X_i - \max_{j \in T} X_j)^+]$ , while  $v(\{i\}) - v(\emptyset) = \mathbb{E}[X_i]$ ; Owen’s identity integrates this over  $T$ . It is payoff-valued and averages over coalitions, unlike the grand-coalition probability  $w_i$ .  $\square$

So there is a clean mathematical link—the winning-probability field integrates along the diagonal to an Aumann–Shapley value of the ability-scaling game—but it should not be read as “approximately Shapley.” The winning probability is a probability share, in different units, and it answers a different question: *selection relevance* (who is the single best in the field) rather than *coalitional cooperation* (average marginal value across coalitions). The sharp way to see the gap: a contributor that is never individually the best but improves every blend it joins—a pure diversifier—carries real coalitional value and earns essentially *no* race credit, because the race only ever rewards being the winner. Where credit means cooperation or combined value, the right object is the Shapley value of a combination-value game, not this. What can be said axiomatically is about the probability share directly, and it should not be dressed in the Shapley axioms (which are conditions on a payoff value of  $v$ ).

**Proposition 2** (Properties of the probability-share credit). *The credit  $w = \nabla G_S(\theta)$  satisfies probabilistic efficiency ( $\sum_i w_i = 1$ ); argmax symmetry (interchangeable contributors receive equal credit); and argmax nullity (a contributor almost surely never the maximizer receives  $w_i = 0$  and leaves the others’ credit unchanged). It is not a linear functional of the game, so it is not additive in the Shapley sense.*

*Proof.* Probabilistic efficiency and  $w_i \geq 0$  are immediate. Argmax symmetry holds because exchanging two contributors with the same marginal and dependence leaves  $G_S$ , hence  $\nabla G_S$ , invariant. For argmax nullity, if  $X_i \leq \max_{j \neq i} X_j$  a.s. then  $i$  is never the argmax, so  $w_i = 0$ , and  $\max_N X = \max_{N \setminus \{i\}} X$  leaves the others unchanged. Non-additivity is direct:  $w = \nabla G_S$  is the gradient of a convex potential, nonlinear in  $\theta$  (strictly convex only on the quotient  $\mathbb{R}^n / \text{span}\{\mathbf{1}\}$ , being affine along  $\mathbf{1}$ ), so the map from game to credit is not linear ( $w(\theta + \theta') \neq w(\theta) + w(\theta')$ —the right-hand side need not even lie on the simplex).  $\square$

These are the right analogues: they are statements about a probability share, not the Shapley efficiency or null-player axioms, which concern a payoff value of  $v$ . Trading the linearity that pins down the Shapley value is what lets the credit be a cheap, differentiable gradient.

## 4 Computational complexity

The practical case for the winning probability as credit is its cost, stated against the right baseline. For a *generic* black-box cooperative game, exact Shapley enumerates  $2^n$  coalitions, and the usual recourse is permutation sampling,  $O(RMn)$  at  $O(1/\sqrt{R})$  error. But the expected-maximum game (5) is not black-box: on an empirical sample  $X^{(1)}, \dots, X^{(M)}$  the max-game Shapley value has order-statistic structure and can be computed in about  $O(Mn \log n)$  without enumerating coalitions. So the race’s edge over a *specialized* max-game Shapley algorithm is not exponential. What the race buys is different: it returns the endpoint-gradient credit—the probability share—in a single  $\arg \max$  pass over the  $M$  draws,  $O(Mn)$ , with no coalition or permutation machinery, and as the gradient of a smooth potential it is differentiable in the abilities and supports cheap online updates. Against a generic Shapley evaluator the gap is the exponential one; against the best specialized one it is the constant factor of a single pass, plus differentiability. The race is *exact for the empirical simulation law* defined by the  $M$  draws; for the population law it carries the usual  $O(M^{-1/2})$  Monte-Carlo error.

## 5 Redundancy, and the calibration condition

The reason to prefer this credit rule over a correlation-blind one is redundancy.

**Proposition 3** (Redundancy consistency at fixed abilities). *If contributors  $i, j$  have equal ability and identical performances ( $X_i = X_j$  a.s.), then either wins exactly when the other does; with symmetric tie-splitting each receives half of the single-contributor winning probability, and the pair’s combined credit equals that of one contributor.*

This is a property of the race *at fixed, equal abilities*. It does not follow for an arbitrary vector of scores fed through  $\arg \max$ : if near-duplicates carry slightly unequal scores, the largest wins almost every draw and the split is winner-take-all. The fix is calibration (3): set the benchmark  $w^0$  to the contributors’ *marginal* credit—equal across genuine duplicates—then tilt under their empirical dependence, so the pair is de-duplicated and split evenly. The price is that the de-duplicated weight redistributes in part to weak contributors; spreading credit and concentrating it cannot both be exact.

## 6 Applications

*Forecast combination.* Supplying the race with the base forecasters’ per-period losses  $L_{k,t}$  (so the lowest-loss forecaster wins) yields combination weights that are a redundancy-aware credit split over models. On the M4 competition this is a promising but inconclusive direction (the companion paper reports that study): in the small-sample regime it is competitive with, and the most stable member of, the combination family, but we claim no out-of-sample superiority. The performance variable must be a loss, not a signed error, or the “winner” is merely the most negatively biased forecaster.

*Feature attribution.* For model interpretability the contributors are features and the credit is a redundancy-aware importance. Raw contribution scores give a quick screen but are winner-take-all among correlated features; the calibrated race delivers the even split that correlated-feature attribution wants, far more cheaply than a coalition-based method. Unlike SHAP it returns a share of wins rather than an additive decomposition of a single prediction—a global importance, not a local explanation.

*Variable selection.* Thresholded, the same importance is a selection rule, and the natural competitor for *stable* selection under collinearity is stability selection (Meinshausen and Bühlmann 2010), which ranks features by lasso selection frequency over subsamples. On collinear designs the calibrated race is the more stable of the two: across data resamples its per-duplicate credit varies several-fold less than both a single lasso (which picks one copy and flips) and stability selection, while splitting redundant features evenly—and it does so in one pass rather than a bag of lasso fits. The costs are a softer signal/noise contrast, since some credit leaks to weak features, and a higher per-call price than a single fit.

## 7 Discussion

The winning probability is a fast, differentiable, online, redundancy-aware credit rule, useful where redundancy-aware credit is wanted and a coalition-by-coalition computation is too expensive. It is *not* a Shapley value, and the distinction is a choice of question, not an approximation: Shapley measures *coalitional fairness*—a contributor’s average marginal value as coalitions are assembled, a counterfactual-historical quantity—while the winning probability measures *field relevance*, how often a contributor is uniquely the best in

the field one actually has. The race does not see coalitional value at all: a pure diversifier, never individually best, earns no race credit however much it improves blends. For legal or moral credit the coalitional average is the right object; for live selection—ensemble weighting, feature screening, routing, “which signal do I rely on now?”—the full-field endpoint question is the operational one, and it is the one the race answers. Its redundancy guarantee is exact for the calibrated equal-ability race, which is the construction to use; raw scores give only a winner-take-all screen. Its axiomatic standing is settled by Proposition 2: it keeps efficiency, symmetry, and the null-player property, trading away only additivity. As a variable-selection rule it is the stable, redundancy-aware option of Section 6. Its choice-sensitivity Jacobian  $\nabla_{\theta} w$  turns out to be well-conditioned even when the covariance is not: numerically its condition number stays near 1 while the covariance’s grows into the hundreds, so the ill-conditioning that defeats  $\Sigma^{-1}$  never enters the choice-space inverse and calibration needs no preconditioning.  $J$  is also informative in its own right: it is a *substitutability matrix*— $J_{ij}$  measures how much contributors  $i, j$  compete for the same winning event—so clustering it recovers the redundancy groups *endogenously*, from the allocation object itself, and in choice geometry rather than raw correlation. Two contributors can be highly correlated yet not substitutes in the race if one dominates and the other never wins; the correlation matrix cannot tell these apart but  $J$  can. We leave this interpretability layer to future work. The cost that does remain—the de-duplicated credit that leaks to weak contributors, the one weakness recurring across the attribution and selection experiments—is controlled by a concentration (inverse-temperature) parameter  $\beta$ : scaling the calibrated strengths by  $\beta$  drives a sub-maximal contributor’s credit to zero as  $\beta \rightarrow \infty$  while leaving the even split among genuine duplicates intact, trading the leak against the smoothness of the map (developed in the companion paper). A complementary device is an *outside option*: a null competitor whose winning probability  $w_0$  absorbs mass no real contributor deserves (an interpretable baseline—unexplained credit, or cash / market-mode in the portfolio reading). Empirically the null preserves the even split and softens the leak with a mild bias toward weak contributors, but it shaves mass roughly in proportion to share, so it is a smooth interpretable floor rather than a targeted leak remover; temperature is the sharper tool. The whole construction is, in one line, a replacement of additive marginal contribution by the probability of being the *non-redundant winner in the actual field*. It is implemented in the open-source `allocation` package, built on `thurstone`.

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